

Table VS2. Virtual screening of the sixty designed compounds with ECFP4-based classifiers.

The same sixty compounds shown in Table VS1 were scored by the saved random forest (RF), XGBoost and multilayer perceptron (MLP) classifiers trained on ECFP4 fingerprints. Values are active-class probabilities expressed as percentages. *Active Hits* counts the classifiers assigning a probability of at least 50% to the active class. At least two active hits constitute a majority-active prediction. Series **IX** was the only series for which all six members met this criterion.

Code	SMILES	RF-ECFP4 (%)	XGBoost-ECFP4 (%)	MLP-ECFP4 (%)	Active Hits
<i>Series I</i>					
Ia	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	25.70	11.15	77.97	1
Ib	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	35.50	9.92	54.90	1
Ic	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	36.50	29.44	85.03	1
Id	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	27.70	11.15	83.34	1
Ie	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	38.50	9.92	62.70	1
If	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	40.00	29.44	88.63	1
<i>Series II</i>					
IIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	36.20	74.30	89.97	2
IIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	41.29	19.78	42.81	0
IIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	36.70	15.47	51.58	1
IId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	38.20	74.30	89.06	2
IIE	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	43.79	19.78	42.41	0
IIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	39.70	15.47	49.90	0
<i>Series III</i>					
IIIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	31.70	10.76	49.55	0
IIIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	33.79	11.84	14.50	0
IIIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	23.70	8.24	8.81	0
IIId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	34.70	10.76	49.18	0
IIIe	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	37.79	11.84	14.62	0
IIIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	27.20	8.24	8.65	0
<i>Series IV</i>					
IVa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	37.70	23.55	71.37	1
IVb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	40.04	31.45	32.20	0
IVc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	33.70	18.66	20.64	0
IVd	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	39.70	23.55	53.76	1
IVe	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	44.04	31.45	17.60	0
IVf	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	36.70	18.66	10.14	0
<i>Series V</i>					
Va	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	44.70	77.82	82.43	2
Vb	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	49.79	87.49	47.25	1
Vc	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	40.20	72.32	31.92	1
Vd	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	46.70	77.82	67.14	2
Ve	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	52.79	87.49	26.16	2
Vf	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	44.20	72.32	15.98	1

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Table VS2 (continued)

Code	SMILES	RF-ECFP4 (%)	XGBoost-ECFP4 (%)	MLP-ECFP4 (%)	Active Hits
<i>Series VI</i>					
VIa	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	47.00	32.04	80.49	1
VIb	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	45.29	39.63	41.91	0
VIc	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	39.00	22.28	27.89	0
VIId	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	48.50	32.04	62.46	1
VIe	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	48.79	39.63	21.38	0
VIIf	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	42.50	22.28	12.92	0
<i>Series VII</i>					
VIIa	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	47.00	52.80	96.07	2
VIIb	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	47.00	66.14	85.07	2
VIIc	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	42.00	62.20	74.12	2
VIIId	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	48.00	52.80	90.79	2
VIIe	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	49.50	66.14	65.71	2
VIIIf	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	44.50	62.20	47.61	1
<i>Series VIII</i>					
VIIIa	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	43.20	60.26	74.04	2
VIIIb	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	43.92	64.77	34.28	1
VIIIc	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	40.70	75.12	20.72	1
VIIId	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	43.20	60.26	49.74	1
VIIIe	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	45.92	64.77	13.93	1
VIIIIf	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	41.70	75.12	7.58	1
<i>Series IX</i>					
IXa	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C32</chem>	54.20	20.17	94.61	2
IXb	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32</chem>	58.00	48.06	85.43	2
IXc	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C32</chem>	52.00	75.86	96.02	3
IXd	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C3N=C2</chem>	54.70	20.17	94.58	2
IXe	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2</chem>	58.50	48.06	85.08	2
IXf	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C3N=C2</chem>	52.50	75.86	95.92	3
<i>Series X</i>					
Xa	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	48.70	42.06	36.00	0
Xb	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	43.20	30.63	4.80	0
Xc	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	49.20	42.06	17.77	0
Xd	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	47.92	46.89	3.31	0
Xe	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	44.70	30.63	1.94	0
Xf	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	45.92	46.89	8.29	0